

Lesson 4 Assignment Reflection

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Reflection

In this assignment, I worked with a simple multi agent system that recommends a supplier for a factory request. So, the system has two main agents and one tool. The buyer agent receives the request from the user and identifies important information such as the quantity, deadline, and any priority at the lowest cost. It then sends this information to the supplier's analysis agent. After the analysis is complete, the buyer's agent receives the recommendation and returns the final response to the user.

The supplier analysis agent is responsible for comparing the available supplier, and it looks at price, delivery time, reliability, and capacity and in the modified version, I made this agent check two important constraints which a supplier must have enough capacity for the requested quantity, and its delivery time must be within the requested deadline. If a supplier fails either of these conditions, it is not considered for the final recommendation. The agent also considers the user's priority as if the user asks for the lowest possible cost, and the system chooses the lowest-cost supplier from the suppliers that are actually feasible.

The supplier information tool provides the supplier data used by the system, and this includes the supplier's name, price, delivery time, reliability, and capacity. The agents should only use the information returned by this tool and during the baseline tests, I noticed that the LLM sometimes stated things that were not supported by the tool data and it incorrectly said that verinox had the lowest price even though its price was \$6.40 and dalby's price was around \$3.85. To me, this showed that the LLM could make incorrect numerical comparisons even when the correct data was available.

The coordination strategy in this system is mostly sequential and orchestrated because the buyer agent starts the workflow, sends the request to the supplier analysis agent, and the supplier analysis agent uses the supplier information tool. The result is then returned to the buyer agent and finally to the user. The agents have a limited level of autonomy. They can also analyze the request, retrieve supplier information, compare suppliers, and make a recommendation without the user controlling every step, and this makes it easy.

The system does not actually place an order or make a real purchase because it only gives a recommendation. This is appropriate because a human can still review the result before making a real decision.

The UML diagrams helped me understand the workflow more clearly. The use case diagram showed the main user, system boundary, agents, tool, and the main functions of the system. The sequence diagram showed the order of communication between the factory user, buyer agent, supplier analysis agent, supplier information tool, and the local LLM. It also helped show what happens when no supplier is feasible.

There are several risks and limitations in this system because first, the LLM can make incorrect comparisons or hallucinate information, and second, the output can vary between runs even when the same request is used. For the last system depends on the supplier's data being correct and up to date.

Another limitation is that the local Qwen model sometimes struggled with numerical reasoning because of this and improving the prompt alone was not enough. I added deterministic checks for capacity, delivery, and supplier facts while keeping the LangChain and Ollama workflow. The system also only uses a small number of supplier attributes and does not connect to real supplier APIs, so it cannot verify real-time prices or availability.